Content and Task Based Navigation for Structural Biology in 3D Environments

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ABSTRACT

Visualisation and exploration of molecular experimental results play a crucial role in structural biology. By visualizing and analysing the 3 dimensional structure of a molecule - possibly over time thanks to simulation tools - scientists try to understand its functional role in the cell. Stereoscopic rendering features are historically used in structural biology and experts are thus quite familiar with virtual environment and 3D interaction. Since the dawn of virtual reality, immersive environments have been used to bring scientists into the heart of complex molecular scenes while adding an interactive dimension. However, in immersive environments as well as in desktop contexts, many issues concerning navigation need to be addressed when exploring molecular content. Among these issues, lack of spatial awareness and the cybersickness phenomenom encountered in navigation tasks are major obstacles to overcome to avoid any degradation of the experience and efficiency. It is interesting to highlight the fact that even if navigation is very frequently studied in virtual environments, most of the studies produce generic paradigms that are only applicable to realistic virtual scenes. These approaches do not explicitly take into account the content of the 3D scene and the task of the end user. In this study, we present some new implementation of navigation paradigms based on tasks and contents, dedicated to molecular biology and designed with the involvement of experts in structural biology. These paradigms are independent of the interaction context and can be indifferently used in a daily desktop context or in immersive environments ranging from CAVEs to HMDs.

Index Terms: Computer Graphics [I.3.7]: Three-Dimensional Graphics and Realism—Virtual Reality

1 INTRODUCTION

Understanding the function of a molecule stands as one of the main challenge of structural biology. One major component of the analysis of a molecule lies in the knowledge resulting from the exploration of its structure. A well-known adage in structural biology states that a protein's shape directly determines its function [19]. Visualization of such structures is therefore a crucial step in the work of structural biologists. Numerous efforts have been made these last two decades to improve the way experts look at molecules. To tackle the issue of dealing with such abstract objects, several representations have been chosen over the years and today the community has broadly adopted some of them [11]. Recently, these representations have undergone evolutions thanks to continuous improvement in the performance of computing systems. Game engines and their rendering algorithms have been used for example to set up new representations of atoms using GPU capabilities, aiming at dynamic representations of large scale molecular data [4]. Recent works still try to find the best compromise in terms of visual data and information load we can couple for molecular visualization [1].

To complement the information carried by the representation, recent studies showed that stereoscopic viewing of molecular structures brings a significant improvement in the way experts perceive the molecules [23, 21]. This improvement through better depth perception of objects allowed by stereoscopic rendering can be explained by the nature itself of the objects we are looking at. First, the data observed in molecular biology are atoms, objects carrying intrinsically 3D information that need some adaptations to be represented in 2D. Moreover, molecular complexes comprise between a few hundred particles to several millions for the largest systems that can be rendered nowadays. Such a large number of particles leads to a loss of spatial landmarks during molecular exploration when performed with a 2D screen. On top of their stereoscopic rendering, immersive environments are a good provider of new ways to interact with the data. Few years after the implementation of the first CAVEs (Computer Assisted Virtual Environments), several studies started to develop immersive tools to process molecular modelling data using both the new display capacities as well as the new ways to interact with data [5, 7]. Haptic devices and the feedback they provide have proved to significantly improve several of the processes used in molecular modelling [17].

Within immersive environments, arises the need for navigation processes that will allow the user to fully enjoy his/her new perception of the data. However, a major bottleneck when exploring a virtual scene in immersive conditions stands in the fact that users can quickly experience various symptoms similar to motion sickness. The so-called *cybersickness* phenomenon can significantly reduce user comfort and efficiency in performing a specific task in a virtual environment [15]. This is particularly important when we focus on the navigation part of any immersive exploration of data. Indeed, given the importance of the user comfort when performing a task that could last for long periods, it is necessary to tackle the issue of sickness when dealing with immersive environments and abstract data together. One factor involving the *cybersickness* relates to the size and density of the virtual environment used in structural biology. Without guides and spatial landmarks, free navigation in a 3D representation of a molecule immersed in a totally empty and uniform universe can significantly impact user comfort.

The need for alternative ways to navigate in virtual environments has been emphasized by Hanson et al. [10] who cited several families of navigation paradigms that significantly increase the experience of the users. Van Dam et al. define the navigation tasks for scientific visualization as being divided into 3 categories [23]:
- **Exploration**, navigation without explicit target
- **Research**, moving through an environment toward a particular location

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- **Manoeuvring**, short and high-precision movements to position the users to perform a task

This perfectly fits the mantra enounced by Shneiderman, which consists in a process where overview is the preliminary step before zooming and filtering to finally, get details on demand [20], and that define roughly the process of data visualization.

We focus here on the importance of navigation in the data visualization process. We want to emphasize the major roles that both content and task play during a virtual exploration session. Over the years, in parallel with the development of immersive environments, new navigation techniques have been developed to optimize the navigation processes in virtual scenes [22, 14, 3]. The video game industry, professional simulators and most of fields working with realistic scenes have optimized their navigation approach to take full advantage of the capabilities of immersive environments. In expert programs dedicated to molecular visualization, very few efforts have been made to take into account the specific features of scene content and task for the process of molecular visualization. The navigation process often includes object manipulation, and the paradigms behind this manipulation have not evolved since decades. Programs like PyMol [6] or VMD [12] base their process for molecular motion through the manipulation of the object adapted to 2D visualization and mouse/keyboard interaction. No adaptation is implemented to take into account the nature of the observed molecule or the task to be performed: users will observe a small peptide of few hundred of atoms in the same way as a virus of several million atoms. There is therefore much room for improvement and a need for new ideas in order to adapt the process of navigation.

In this paper, we show how we addressed each of the cited navigation steps involved in the study of molecular complexes. We will start with a short description of the material coming from preliminary ergonomic studies. We will emphasize several developments we made, from the external exploration of molecular complexes to the processing of more precise tasks via the use of specific navigation paradigms while using the structure itself of molecular complexes. More precisely, we demonstrate that it is possible to combine structural information to support the navigation process via the symmetrical arrangement present in molecular complexes and their hierarchical architecture. We identified several tasks that structural biologists perform on a daily basis to create navigation paradigms that will help them to achieve their objectives. Finally we will discuss the future improvements that structural biology could get from a more systematic collaboration between ergonomic experts and scientists to setup new visualization processes, through navigation or not.

2 Ergonomic studies and user needs analysis

It has long been acknowledged that system design should rely on knowledge of future users' needs and activities, both to produce and to assess design solutions [18] with the purpose of improving both performance and satisfaction in the use of computerized systems. In the case of designing virtual environment systems, Gabbard and his colleagues [2] has identified four levels of contribution of ergonomics to the design process: user task analysis, expert-guidelines based evaluation, formative evaluation and summative evaluation. User task analysis is typically performed in the early stages of the design process, where it allows producing knowledge about user needs and requirements based on observations of, and interviews with representatives of future users. Few studies already showed the importance of ergonomic studies ahead of specific developments in order to fit and identify the specialists needs and feelings [8].

Following the same line of investigation, we have worked together with experts in ergonomics and with structural biologists in order to analyse the navigation task as it is usually performed, and to identify the features that might lack in the software they use every day. Through a complete analysis of a variety of interviews we were able to highlight several needs for improvement at different steps of the navigation. These needs directly drove our development process by highlighting the features

3 Navigation paradigms

Throughout the exploration process of a molecular complex, navigation is a crucial element that needs to be perfectly mastered by the user to focus only on the analytical part of his work. Standard ways to navigate into molecular complexes using desktop solutions are close to object manipulation. However, in 3D, we are focusing on camera motion when we are navigating in a virtual scene. The user must consider him/herself as if (s)he was in a virtual vehicle where the viewing possibilities are extended compared to usual displays. From this point of view, zooming in or out becomes a movement of the camera for instance. Together with the stereoscopic capability and large display capabilities that are present in most immersive environments, they provide new ways to interact with the virtual scene. A choice must then be made to favour interactions that suit the environment. Navigation plays an important part in the interaction process, but because it is more a way of accessing information than a way of providing it, the navigation task has to be optimized to be as light and direct as possible for the user. By making the navigation process intuitive for the user, we can shift the workload on specific tasks that naturally require most of the user focus.

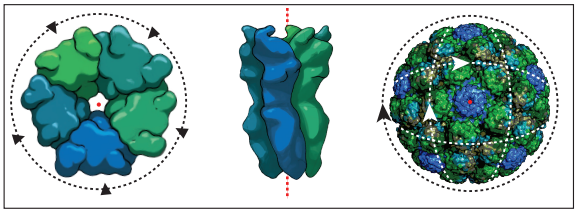


Figure 1: Several types of symmetries found in molecular complexes. On the left/middle, GLIC, a transmembrane protein composed of 5 elements with a single symmetry axis with a top-view (left) and a side-view (right). On the right, a virus capsid with four types of components presenting a symmetry centre overlaying the centre of gravity of the virus.

One means to make the navigation process intuitive to the user is to make it coherent with the data (s)he observes. If the trajectory followed by a user during the exploration has some echoes in the nature itself of what is seen, then we can provide information while the user is performing a basic navigation task. Molecular complexes present a structural arrangement around axes and centre of symmetries that can be used to drive the setup of such *intelligent* paths of navigation. This symmetry feature not only plays a role in the structure of a complex but also in its function [9]. Some examples of such symmetrical arrangements found in biological complexes are illustrated in Figure 1. Either of these examples plays a crucial role in some of the most important metabolic processes taking place in the cell.

At the beginning of an exploration session, the symmetrical axes are either automatically computed or manually provided. They are then used to re-orient the virtual scene toward where the user will progress and act as the basis for the automatic generation of navigation paths. Navigation guides are not limited to the only symmetrical molecules, it is possible to identify principal axes in a protein in order to provide a base for the navigation path generation. The axes identified by the scientists as important are then used as symmetrical axes would be used all along the navigation. In the latter case without clear symmetry within the complex, experts have then the opportunity to choose a preferential direction according to the

protein environments such as membrane planes to re-orient the protein.

One main contribution resulting from the reorientation of the virtual scene is the merging of the main symmetry axis with the up axis of the orthonormal reference frame. By fitting these two main axes, we provide a first spatial landmark to the user. Indeed, all along her/his exploration and as often as it is possible, we will maintain the molecular complex in this fixed orientation. This feature mimics the behavior of any external exploration of a tower by a helicopter for instance, always maintaining the user well oriented compared to her/his object of interest. To go further, we went beyond the usual uniform and monocolor visual environment that use molecular viewers and we added a skybox surrounding the protein, to favour the orientation capability of the users, fitting the guidelines given in [24]. The selected skybox must feature a distinct top and down part that would correspond to the bottom and top of the molecule after its reorientation. As often as the user might want it, it is then possible to get basic but useful information about his orientation in the scene, in a very short time. Experts of the domain usually choose a preferential direction according to the protein environment such as membrane planes. These specific orientations can be rendered, without an explicit model of the membrane but only thanks to the added skybox.

Many insights can be deduced from external views of a molecular complex. We know that the overall shape of a protein can provide significant clues on its functional state. But other information like polarity, accessibility or environment integration are some basic but nevertheless important elements that can be easily guessed from an external exploration of a complex. The first path of navigation when entering the scene is a circular path that consists in an axial rotation around the complex while the camera is always facing the complex. The height of the camera is controllable by the user and the vertical moving axis is kept perpendicular to the symmetry axis. If the user, during his exploration, reaches an external part of the complex, the path is recalculated to keep the focus on the upper or bottom part of the complex while following the global vector model [13]. This model of vectors keeps the camera up as close as possible to a parallel of the symmetry axis and recalculates the orientation to make a smooth transition toward a perpendicular view of the complex when reaching one of the extremities of a complex as illustrated on the Figure 2. The user is allowed to navigate with 3 degrees of freedom in terms of translation. In terms of rotation, the only possible degree of freedom is the one associated with the head-tracking system but no specific rotation via buttons or analogue commands are implemented in this constrained mode of navigation.

3.1 Exploration

The transition between the external and internal exploration of the complex is an important question that cannot be answered by a straightforward algorithm. Some molecular structures present pores or centred structures with an important role in their function. Among them, allowing the passage of ions or water, or bonding with ligands are some functions that can take place buried in the middle of a structure. Given their importance it was necessary to think about efficient paradigms to navigate in these areas as easily as possible while keeping the possibility to perform complicated tasks. To do so, we allow automatic spatial transitions between the inside and outside of the structure by again using the previously detected symmetry axes. Following a user request, navigation paths are calculated to reach the inner part of a protein - during his/her external exploration - to focus on a specific point. When the centre of the complex is reached, a new set of navigation paths is proposed to climb down or up the complex along the symmetry axis. In this mode, only one degree of freedom is kept for the translation and the rotation at the same time. However, it is possible to switch the axis

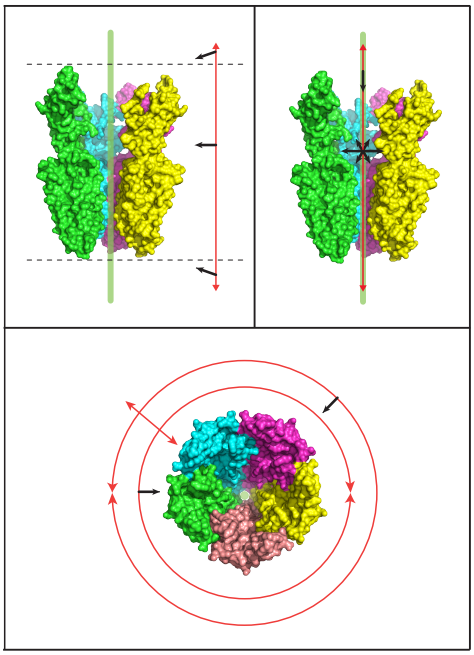


Figure 2: Depending on camera position, different degrees of freedom are offered to the user in terms of rotation / focus of the camera (black arrows) and/or translation / movement of the camera (red arrows). In the top left, vertical and external exploration, users can go up and down, remaining parallel to the symmetry axis. On the top right, vertical and internal exploration, users can go up and down along the symmetry axis by keeping the *up* vector of the camera either parallel or perpendicular to the axis. At the bottom, horizontal and external exploration, user can go right and left by following a circle path and/or zooming in/out of the complex.

on which the degree of freedom is applied in order to mimic the behaviour of a panoramic lift for instance or the passage through a subway path.

Symmetry brings new ways to compare repeated regions of interest. When a particular event occurs in a specific area, it is not rare to see it repeated on another subunit because of the structural similarities of the two regions. We use the symmetrical link that binds the subunits of a multimer together in order to set up a quick switching mode allowing users to *jump* from one subunit to the other while keeping the focus of the camera on the specific area. The user can then keep track of an event somewhere and compare it with the similar ones occurring on repeated regions elsewhere in the structure. The transition is quick and completely automatic.

3.2 Finding an optimal point-of-view

Even if the exploration can bring a large amount of information, several molecular events concern only few atoms or residues. In large structures comprising several millions of particles, these areas can become complicated to visualize because of the density of the surrounding particles. Moreover, certain areas are deeply embedded in the molecular complex, transforming the simple task of visualization into a complex challenge for the user. To tackle this issue, we developed an algorithm that is able to provide the best point-of-view for the camera, given a target and a camera distance. This algorithm takes into account the neighbouring atoms of a given target and computes the largest cones of view. The user will provide the cones height and will be able to choose the view that fits the most his wishes.

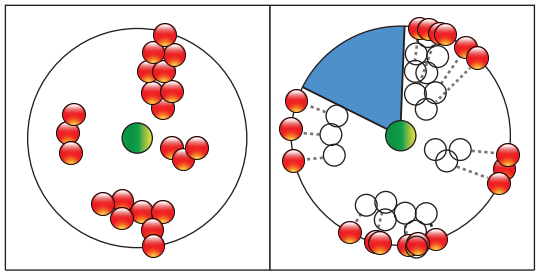


Figure 3: on the left, our target is represented in green and surrounded by several atoms in red. The circle in black represents the distance between the target and the camera putative best position. On the right, illustration of the neighbours projection on the sphere when we transform their Cartesian coordinates into spherical ones.

3.2.1 Algorithm

The algorithm we use to compute the cones of view can be decomposed as follows: 1. The user provides the target coordinates either in a raw format (x, y, z) or via a selection from which the centre of mass coordinates will be calculated. A cone of view of height d is taken as input and will be considered as the camera distance to the target.
2. All atoms located within the sphere of radius d and centred at the target coordinates are considered as neighbours and putative occlusions for the cone of view. Their 3D Cartesian coordinates are stored.
3. The Cartesian coordinates of the neighbours are shifted to consider the target coordinates as the origin of the new reference frame. This allows the creation of a spherical reference frame centred on the target. Then, all coordinates are transformed into spherical coordinates to match the new reference frame. A spherical coordinate is composed of three parameters: radial distance to the origin r, a polar angle θ (theta) and an azimuthal angle ϕ (phi). By putting aside the radial distance, we only get the direction of the atom with respect to the reference frame origin, here the targeted atom. By fixing the radius of all neighbours to r, we can see this transformation as a projection of all atoms on a sphere of radius r. The idea is then to find the biggest empty circle on the sphere surface that will correspond to the largest cone with no atom inside. We illustrate the concept in the Figure 3.
4. Since r is fixed, phi and theta angles of the neighbours are then plotted in a 2D matrix. The resulting plot renders the flattening of the sphere in two dimensions. The matrix is also extended of 50% along x and y to take into account the periodicity of the sphere (the maximum size of an empty circle being of the size of the matrix longest side, no more than 50% extension is needed). Indeed, if no extension is done, we could miss solutions that would appear on the extremity of the matrix, where the two parts of the flattened sphere are joining each other. As we can see on the Figure 4, without this extension, we would have missed 2 of the largest empty circles. The plotting can be seen as a 2D map of the neighbours distribution around the sphere following their projection on it.
5. From the list of the phi/theta pairs, a Voronoi diagram is computed. We get a list of Voronoi vertices, each of them associated to a triplet of the closest points.
6. By calculating the distance between each Voronoi vertex and the closest three points, we search for the largest empty circle. This circle will be centred on the vertex and will have the radius of the distance between the vertex and its points. We recall that in the Voronoi algorithm, each vertex is the circumcentre of a set of at least 3 points, the closest points from the vertex among the whole list of points considered. The circle with the largest radius will be considered as the largest empty circle (Figure 4). It is important to note that, despite the extension of the matrix, we do not select circle centres that lie outside the box surrounding the first set of atoms spherical coordinates (blue points on the Figure 4). However, the points that belong to the circle can be in the extension part of the plot.
7. The phi/theta values of the largest empty circles centres are transformed back to Cartesian coordinates then re-shifted to fit the default reference frame. The radius of the circles will give us an aperture angle that will be used later to compute an optimal trajectory. We will explain this part below.
8. These coordinates can then be iteratively chosen by the user to place the camera. The camera will always face the target.

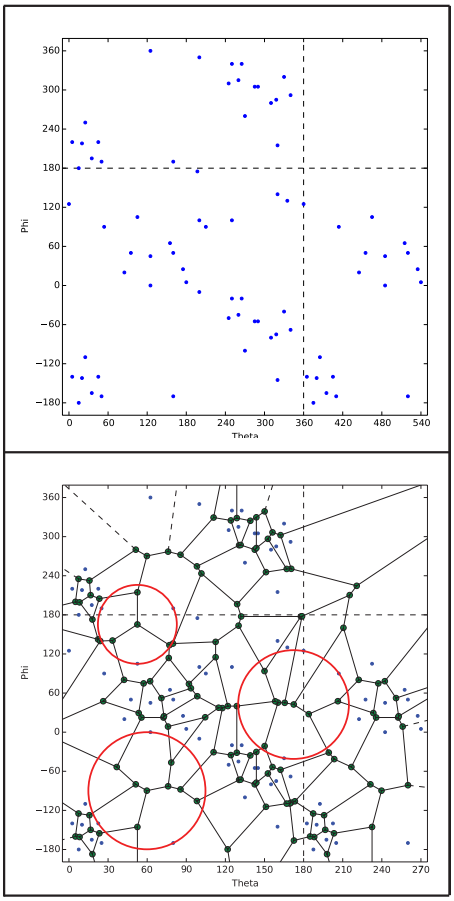


Figure 4: On the top, plot of phi against theta angles of neighbours atoms surrounding the target. Each atom is represented by a blue dot. At the bottom, a plot with vertices and segments got from Voronoi tessellation represented as a Voronoi diagram. Green dots represent Voronoi vertices and black lines represent Voronoi segments. Red circles are the largest empty circles that can be found.

This algorithm has a low complexity θ(n log n), with n standing as the number of atoms) since the research of our target neighbors is computed with n-1 iterations (constant) and the voronoi diagram with the research of largest empty circle is computed in θ(n log n) in the worst case where all atoms are considered as neighbors of the target. It is able to provide a list of the most cleared views over a target nearly instantaneously. It allows the user to switch between several optimal viewpoints, or to compute in interactive time camera paths from the current view to the optimal viewpoints as explained in the next section.

3.2.2 Camera transition

In addition to the best point-of-view paradigm, we added a smooth camera transition between the user current camera position and the new one computed by the algorithm. This transition is made to let the user evaluate the location of its target and to preserve his awareness of his target environment. An automatic and progressive approach is setup to drive him from his position to the one defined in the output of the algorithm. To compute this transition, we set the focus of the camera on the target while an optimal trajectory is followed. A part of this trajectory is computed by repeating the algorithm described previously with a larger camera distance r and restrained to an aperture angle that corresponds to the first apertures we got during the first iteration of the algorithm. As we can see on the Figure 5, we can obtain a trajectory with the least occlusions all along the approach of the target. By offering different ways to access his target, the user can decide to follow a trajectory made of the longest cleared path instead of the largest empty cone of view. Depending on the distance between the camera position when the algorithm starts and the optimal position, part of the movement might be computed as a straight line to reduce the number of largest empty circle algorithm calculations that may otherwise become too important and costly. The strategy to go from the current camera location to the next optimal point-of-view is based on classical interpolation techniques with respect to the forward constraints on target and the up axis constraints according to the symmetry.

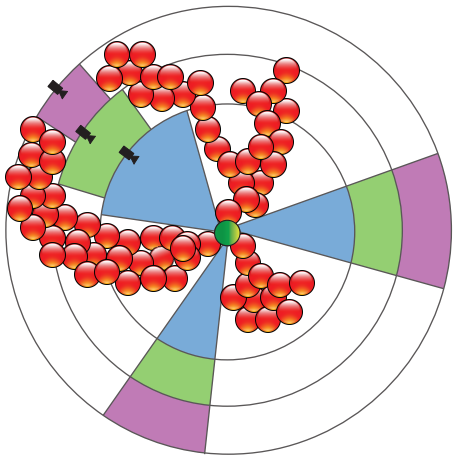


Figure 5: Schematic representation of the best point-of-view algorithm and the computation of the camera path. The target is represented in green and its surrounding atoms in red. Each circle represents a distance between the camera and the target. For each of these distances, the algorithm of best point-of view is computed in order to get the optimal view over the target. The blue cones represent the largest cones of view for the distance provided as input. In light green and purple, we represent the largest empty areas between the previous distance taken as input and the current one. The camera will follow the symbols added on top of the circle to keep its field of view as empty as possible until reaching the final position (at the edge of the blue circle).

In the case where the area around the target is really dense in terms of the number of atoms, we added the possibility to render any atoms that would be between the camera and the target as transparent. This rendering is not done all over the frustum view but within a circle of a fixed size centred on the centre of the screen. This will avoid any occlusions along the movement process to reach the best point-of-view position.

3.3 Adaptative exploded view

In some cases, the computation of the best point-of-view may not be enough to obtain a clear view of a deeply buried region or atom. We therefore decided to go further in the visualization process by allowing the user to modify the 3D structure of the molecular complex. This modification must be done intelligently in order to preserve the biological meaning of what is seen by the user. Once again, we based this feature on the symmetrical and hierarchical layout of molecular complexes. Since this symmetry defines the subunits of a complex, we can easily cut apart the whole structure in order to move each subunit in a precise direction without loosing its overall shape (Figure 6). Then, the user has the ability, at any moment, to spread, narrow or simply translate the complex subunits with respect to the symmetry axis/centre of his/her choosing. In a more automatic way, we added the possibility for the complex to be deformed depending on the camera distance to the complex. In this mode, the spreading distance will increase with the user proximity to the molecule. A maximum spreading distance can be defined to allow a precise investigation of a particular area. Thanks to this structural adaptation, interfaces between subunits as well as buried regions can then be revealed. To avoid any unnecessary overload in the process, the default configuration can be reached at any instant.

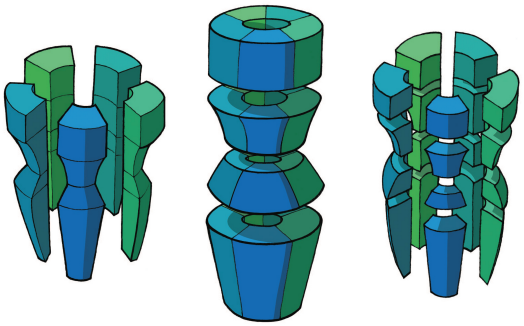


Figure 6: Illustrative rendering of a pore protein following different spreading transformations. On the left, the subunits/chains of the complex have been spread around the symmetry axis of the complex. On the middle, we identify interesting areas along each chain and we spread the protein vertically, in parallel to the symmetry axis. On the right, we combine the separation of the chains and the separation of the remarkable areas identified previously. Illustration courtesy of Davide Spalvieri

4 Conclusion

Data visualization is an analytical process where observation and exploration of data bring information to the scientist. This information might be acquired from an active process from the scientist, by measuring the distance between two objects for instance, or in a passive way, by moving and observing events that cannot easily be described via raw values. As an example, the structural shape of partner binding sites on a protein represents crucial information for the biologist but is complicated to decipher using only information related to volume and polarity. The way to get the information go through a navigation pipeline that drives the user to the event (s)he wants to observe and (s)he wants to perform actions on. The navigation process is then indispensable and becomes of first importance when a user is immersed in his virtual scene. The spatial awareness of a user can be disrupted when (s)he progresses in a virtual scene where no concrete objects can be used as spatial landmarks. To help the user in the navigation process, we provide paradigms and algorithms that are directly based on the content observed and the task performed. By using the structural information

from a molecular complex, our approach is able to generate automatically the optimal navigation paths that favour the complex discovery and analysis. We focused on each of the steps found in the visualization process. We first addressed the exploration of data by creating paths around the molecular complex with transitions for fast exploration of the inside if needed. These paths are made in a way that the user will keep, throughout his/her progress, knowledge of where (s)he goes, where (s)he comes from and from where (s)he passes by. It is important to offer to the user the best possible level of comfort during exploration tasks.

A next step in our approach might be to add a higher degree of interaction with the user to make him choose patterns or structures that will define paths of navigation. For the moment, we remain in a semi-automatic navigation where the user will have a limited set of possible transitions between the modes of navigation offered. However, we will have to keep in mind that any higher degree of interaction and choices offered to a user will increase the cognitive load of the user dedicated to the navigation. And even if studies showed that this increase in the cognitive load is a good way to reduce *cybersickness*, it might also reduce the efficiency of specific tasks other than the navigation.

Feedback from software users is important and used in most software development pipelines. However, it is often forgotten when it comes to post-development efforts. With the evolution of visualization techniques, platforms and approaches, it is important to review the new needs of specialists and the new possibilities offered by the latest technologies. Ergonomic methods will be expected both (a) to allow a clear evaluation of the effects of the navigation techniques described here on the activity of end users, and (b) to provide new insights into their activity and foster the development of new techniques for visualizing and/or navigating computational models of biomolecules. We are still in the process of fully understanding and adapting what biologists consider as important in the environments where the new paradigms might see an evolution thanks to our work.

In order to test and evaluate our algorithms, we implemented them in expert programs of molecular visualisation commonly used by end-users. At this date, the algorithms can be used in PyMol [6] and UnityMol [16] running on MacOSX and Linux for both of them.

It is important to emphasize that our paradigms rest on the content and tasks identified but do not depend on the context. Indeed, these paradigms do not require any specific setup to be fully operational and might be used from standard desktop configuration to immersive environments. Both display technologies and interactive devices do not interfere or impact our algorithms. The genericity of our approach can also be emphasized by the range of its application. Applications in fields dealing with abstract data and presenting a great density of information might benefit from our study. Fluid mechanics, materials science, climatology, medicine or astronomy are all fields where data are not specifically oriented but where visualization plays a crucial role.

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